

4.2. Dada la siguiente historia en un planificador con timestamp:

$H_1 = st_1; st_2; r_2(X); st_3; st_4; r_1(Y); r_4(Z); w_3(X); w_3(Y); w_4(Z); w_2(X); w_1(Y); r_3(Z)$

en donde los valores iniciales de $X; Y; Z$ son 0 y pasa que:

T_1 escribe $Y = 1$

T_2 escribe $X = 2$

T_3 escribe $X = 3; Y = 30$

T_4 escribe $Z = 4$

- Decir que pasa en cada acción y que valores quedan en $X; Y; Z$ si el planificador *no usa multiversión*
- Decir que pasa en cada acción y que valores se tendrían para $X; Y; Z$ si el planificador *usa multiversión MVTO*
- Asumiendo un planificador multiversión, realizar el grafo de conflictos multiversión (GCM) e indicar si la historia MCSR

2) Tiempo	T_1	T_2	T_3	T_4	$WT(X) = WT(Y) = WT(Z) = 0$ $RT(X) = RT(Y) = RT(Z) = 0$ $X = Y = Z = 0$
1	$ST_1 = 100$				
2		$ST_2 = 200$			
3		$r_2(X)$ 0			$TS(T_2) \geq WT(X)$ $200 \geq 0$ $RT(X) = 200$
4			$ST_3 = 300$		
5				$ST_4 = 400$	
6	$r_1(Y)$ 0				$TS(T_1) \geq WT(Y)$ $100 \geq 0$ $RT(Y) = 100$
7				$r_4(Z)$ 0	$TS(T_4) \geq WT(Z)$ $400 \geq 0$ $RT(Z) = 400$

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 $w_3(x)$
 $\overset{3}{\underset{3}{\parallel}}$
 $TS(T_3) \geq RT(x)$
 $\overset{300}{\underset{300}{\parallel}} \quad \overset{200}{\underset{200}{\parallel}}$
 $\wedge$
 $TS(T_3) \geq WT(x)$
 $\overset{300}{\underset{300}{\parallel}} \quad \overset{0}{\underset{0}{\parallel}}$
 $x = 3$
 $WT(x) = 300$
 $C(x) = \text{False}$

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 $w_3(y)$
 $\overset{30}{\underset{30}{\parallel}}$
 $TS(T_3) \geq RT(y)$
 $\overset{300}{\underset{300}{\parallel}} \quad \overset{100}{\underset{100}{\parallel}}$
 $\wedge$
 $TS(T_3) \geq WT(y)$
 $\overset{300}{\underset{300}{\parallel}} \quad \overset{0}{\underset{0}{\parallel}}$
 $y = 30$
 $WT(y) = 300$
 $C(y) = \text{False}$

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 $w_4(z)$
 $\overset{4}{\underset{4}{\parallel}}$
 $TS(T_4) \geq RT(z)$
 $\overset{400}{\underset{400}{\parallel}} \quad \overset{400}{\underset{400}{\parallel}}$
 $\wedge$
 $TS(T_4) \geq WT(z)$
 $\overset{400}{\underset{400}{\parallel}} \quad \overset{0}{\underset{0}{\parallel}}$
 $z = 4$
 $WT(z) = 400$
 $C(z) = \text{False}$

11		$w_2(x)$ " 2	$TS(T_2) \geq RT(x)$ 200 " 200 $TS(T_2) < WT(x)$ <u>DETERMINA T_2</u> <u>HOSTE QUE</u> <u>$C_3(x) = TRUE$</u>
12	$w_1(y)$ " 1		$TS(T_1) \geq RT(y)$ 100 " 100 $TS(T_1) \leq WT(y)$ 100 " 300 <u>DETERMINA</u> <u>T_1 HOSTE</u> <u>QUE $C_3(y) = TRUE$</u>
13		$r_3(z)$ abort T_3	$TS(T_3) < WT(z)$ 300 " 400 <u>ABORT T_3</u> <u>$WT(x) = 0$</u> <u>$WT(y) = 0$</u>
14		c_2	$WT(x) = 200$ <u>$x = 2$</u>
15	c_1		$WT(y) = 100$ <u>$y = 1$</u>
			G_4 <u>$z = 4$</u>

16			ST_3 "500	
17			$w_3(x)$ "3	
18			$w_3(y)$ "30	
19			$f_3(z)$ "4	
20			c_3	

Como T_3 se veche a longer
funden los valores

$$x = 3$$

$$y = 30$$

$$z = 4$$

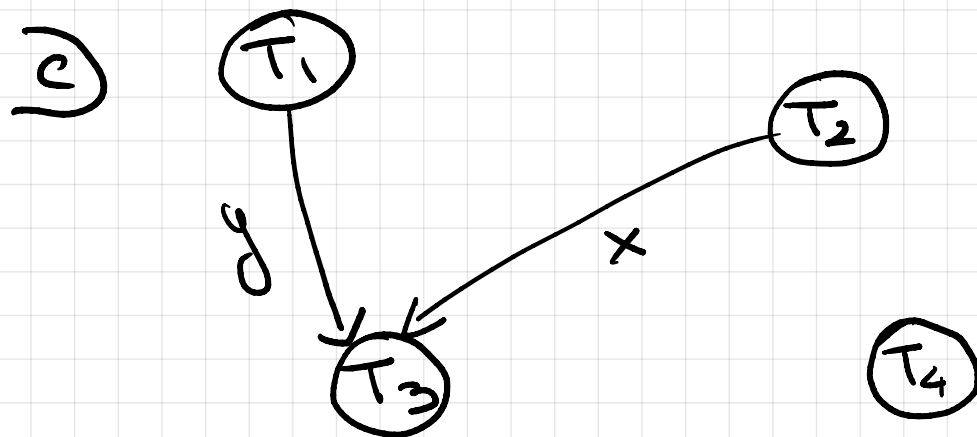
b) πντο

Τιμή	T_1	T_2	T_3	T_4	$x=0, y=0, z=0$
1	$ST_1=100$				
2		$ST_2=200$			
3		$r_2(x)$ "0			
4			$ST_3=300$		
5				$ST_4=400$	
6	$r_1(y)$ "0				
7				$r_4(z)=0$	
8			$w_3(x)$ "3 $x_3=3$		$\neq r_3 /$ $TS(T_k) < TS(T_3) < TS(T_4)$ "0 "300 "200 $x_3=3$ ✓
9			$w_3(y)$ $y_3=30$		$\neq r_4 /$ $y_3=30$
10				$w_4(z)$ "4 $z_4=4$	$\text{Vedie } \neq T_4 \text{ read}(z)$ $z_4=4$
11		$w_2(x)$ "2 $x_2=2$			$x_2=2$
12	$w_1(y)$				$y_1=1$
13			$r_3(z)$ "1 z_0		

Final: $x = 3$
 $y = 30$
 $z = 4$

$i = 1$
 $k = 3$

$T_1 \rightarrow T_3$ si
 $r_1(x_1) \neq w_3(x_3)$ /
 $r_1(x_1) < w_3(x_3)$



$T_i \rightarrow T_k$

si existen pares

$r_i(x_i)$
 $w_k(x_k)$

$r_i(x_i) < w_k(x_k)$

Tiempo	T_1	T_2	T_3	T_4	$x=0, y=0, z=0$
1	$ST_1=100$				
2		$ST_2=200$			
3		$r_2(x)$ 0			
4			$ST_3=300$		
5				$ST_4=400$	
6	$r_1(y)$ 0				
7				$r_4(z)=0$	
8			$w_3(x)$ 0 $x_3=3$		$\neq r_1$ / $TS(T_k) < TS(T_3) < TS(T_1)$ 0 300 100 $x_3=3$
9			$w_3(y)$ 0 $y_3=30$		$\neq r_2$ / $y_3=30$
10				$w_4(z)$ 0 $z_4=4$	Verde $\neq T_4$ read(z) $z_4=4$
11		$w_2(x)$ 0 $x_2=2$			$x_2=2$
12	$w_1(y)$				$y_1=1$
13			$r_3(z)$ 0 $z_3=0$		

pase misero
 dato x

No hay ciclos
 es MCSR